

Center-symmetry confinement, Polyakov-loop order parameter, and the Walker–Wang transport correctness-push: a formalized Svetitsky–Yaffe spine

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SK-EFT Hawking Project (Phase 6d Wave 1)

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We formalize confinement as $\mathbb{Z}_N$ 1-form center-symmetry unbreaking in Lean 4.29, treating the Polyakov loop P as the complex order parameter. The $\mathbb{Z}_N$ center acts on P by multiplication by $\zeta_N = \exp(2\pi i/N)$; we prove this action is a primitive N -th root of unity, fixes only $P = 0$ (confining) when $\zeta_N \neq 1$, and reduces $\zeta_2 = -1$ exactly. Svetitsky–Yaffe universality (D -dim deconfinement = $(D-1)$ -dim $\mathbb{Z}_N$ spin model) is encoded as a structural map: $\text{SU}(2) \rightarrow \text{Ising}$ ($\nu_{\text{Ising}} = 0.6299$), $\text{SU}(3) \rightarrow \text{3-state Potts}$ ($\nu_{\text{Potts}} = 0.5$). The KSS bound $\eta/s \geq 1/(4\pi) \approx 0.0796$ is pinned to a tight bracket $[0.07, 0.08]$ via Mathlib π -bound lemmas, and the Walker–Wang transport prediction — the Phase 6d.1 correctness-push, HPC-gated for numerical validation in the Phase 6B HPC roadmap — is encoded as a tracked hypothesis $H_{\text{WW}} : \eta/s \in [\text{KSS}, 2\text{KSS}]$ with two concrete numerical falsifiers ($\eta/s = 0$ and $\eta/s = 1$). The module ships 18 substantive Lean theorems with 0 sorry and 0 new axioms, a Python mirror with 27 pytest cross-checks (27/27 pass), and two substantive cross-bridges: a genuine biconditional `higher_form_discrete_iff_non_abelian` (not an identity wrapper of the upstream gauge-erasure theorem), and a card-of-fusion-ring bridge to the $\text{SU}(3)_1$ fusion category in `SU3kFusion.lean`. Three post-wave-strengthening corrections were applied under the ruthless review pass (Section ??); the discipline reduced first-pass strengthening to a single ruthless-review iteration.

I. MOTIVATION AND SCOPE

The standard QCD-confinement framing as a Wilson-loop area law admits an alternative, structurally cleaner formulation: confinement as $\mathbb{Z}_N$ 1-form center symmetry unbreaking, with the Polyakov loop as the order parameter and the deconfinement transition matched to a $(D-1)$ -dim $\mathbb{Z}_N$ spin model via Svetitsky–Yaffe universality [?]. The framing connects naturally to the project’s existing higher-form symmetry infrastructure in `GaugeErasure.lean` [?], which already classifies the $\text{SU}(N)$ continuous gauge sector as producing a *discrete* 1-form symmetry — but stops short of identifying the discrete symmetry as $\mathbb{Z}_N$.

This Wave 1 supplies that identification, the Polyakov-loop order parameter, the Svetitsky–Yaffe map, the Kovtun–Son–Starinets (KSS) viscosity lower bound [?], and the Walker–Wang anyon-mediated transport prediction [?] — the project’s 6d.1 correctness-push (analytical only; HPC-gated numerical validation is in Phase 6B).

II. CENTER SYMMETRY, POLYAKOV LOOP, AND CONFINEMENT

A. The cyclic center $\mathbb{Z}_N$

The cyclic center $\mathbb{Z}_N$ for $\text{SU}(N)$ is encoded as a Lean structure `CenterZN` with a load-bearing constraint $N \geq 2$ (the $N = 1$ case is trivial). The fundamental phase $\zeta_N = \exp(2\pi i/N)$ generates $\mathbb{Z}_N \subset \mathbb{C}^*$. Two algebraic properties are proved:

$$\zeta_N^N = 1, \quad (1)$$

$$|\zeta_N| = 1. \quad (2)$$

Equation (??) (`centerPhase_pow_N`) follows from `Complex.exp_nat_mul` and `Complex.exp_eq_one_iff`; Equation (??) (`centerPhase_norm_one`) follows from `Complex.norm_exp` and the vanishing real part of $2\pi i/N$. A concrete computation specializes $\zeta_2 = -1$ via `Complex.exp_pi_mul_I` (`centerPhase_Z2_eq_neg_one`) — the $\text{SU}(2)$ center is the multiplicative group $\{+1, -1\}$ acting on the Polyakov loop.

B. Polyakov loop and the confining predicate

The Polyakov loop is encoded as `abbrev PolyakovLoop : Type := ℂ`, with

$$\text{Confining } P \iff P = 0, \quad (3)$$

and the center action `centerAction` $Z P := \zeta_N \cdot P$. Two substantive theorems link the order parameter to the symmetry-breaking structure:

- `confining_invariant_under_center_action`: $P = 0 \Rightarrow \zeta_N \cdot P = P$ (confining configurations are center-invariant).
- `nonzero_breaks_center_invariance`: $P \neq 0 \wedge \zeta_N \neq 1 \Rightarrow \zeta_N \cdot P \neq P$ (non-zero Polyakov loops are NOT center-fixed).

Together these formalize the order-parameter $\leftrightarrow$ symmetry-breaking link: $P = 0$ is the unique fixed point of the (non-trivial) $\mathbb{Z}_N$ center action on $\mathbb{C}$.

C. Magnitude as a real-valued order parameter

The conventional lattice order parameter is the modulus $|P|$. We prove `confining_iff_magnitude_zero`, the qualitative- $\leftrightarrow$ -quantitative link

$$\text{Confining } P \iff |P| = 0, \quad (4)$$

and the strict-positivity statement `deconfining_implies_magnitude_positive`: $\neg \text{Confining } P \Rightarrow 0 < |P|$.

III. SVETITSKY–YAFFE UNIVERSALITY

The Svetitsky–Yaffe map is encoded as a function `svetitskyYaffeClass : CenterZN  $\rightarrow$  UniversalityClass`, with concrete cases $\text{SU}(2) \rightarrow \text{Ising}$ and $\text{SU}(3) \rightarrow \text{3-state Potts}$ (per Refs. [? ?]). The 3D correlation-length critical exponents are anchored to literature:

$$\nu_{\text{Ising}} = 0.6299, \quad \nu_{\text{Potts}} = 0.5, \quad (5)$$

where ν_{Ising} tracks the Pelissetto–Vicari and Kos–Poland–Simmons–Duffin conformal-bootstrap value [?] and $\nu_{\text{Potts}} \approx 1/2$ reflects the mean-field-like / weakly-first-order nature of 3D 3-state Potts.

The substantive separation between Z_2 and Z_3 deconfinement is formalized as the direct comparison

$$\nu_{\text{Potts}} < \nu_{\text{Ising}} \quad (6)$$

(`ising_nu_gt_potts_nu`, a strengthening-pass result replacing the somewhat-arbitrary 0.6 threshold of the first-pass theorems `ising_nu_above_0_6` and `potts_nu_below_0_6`). The pair of threshold theorems is preserved as quantitative literature anchors; the direct comparison is the load-bearing physics statement.

Figure ?? (left) visualizes the $|P|$ vs T/T_c behavior across the Z_2 and Z_3 universality classes via a schematic $|P| \propto (T/T_c - 1)^\beta$ scaling ($\beta_{\text{Ising}} = 0.326$, $\beta_{\text{Potts}} = 0.111$).

IV. KSS BOUND AND THE WALKER–WANG TRANSPORT CORRECTNESS-PUSH

The Kovtun–Son–Starinets bound on shear viscosity to entropy density is [?]

$$\eta/s \geq \text{KSS} := \frac{1}{4\pi}. \quad (7)$$

We prove a triple of statements pinning KSS to a tight literature-anchored bracket:

1. `KSS_bound_positive`: $0 < \text{KSS}$.
2. `KSS_bound_below_0_08`: $\text{KSS} < 0.08$.
3. `KSS_bound_above_0_07`: $0.07 < \text{KSS}$.

Statements (2) and (3) use Mathlib π -bound lemmas `Real.pi_gt_d4` and `Real.pi_lt_d4`, giving $0.07 < 1/(4\pi) < 0.08$ as a true numerical bracket (NOT a within-own-band tautology in the Phase 6b.1 sense — the bracket width is $\sim 12\%$ of the value, not $\pm 2\sigma$).

The Walker–Wang anyon-mediated transport prediction is encoded as a tracked hypothesis with two independent constraints:

$$H_{\text{WW}}(\eta/s) := (\text{KSS} \leq \eta/s) \wedge (\eta/s \leq 2 \text{KSS}). \quad (8)$$

The two conjuncts pass the drop-conjunct test (each can fail independently). The lower bound is the universal KSS constraint; the upper bound is the Walker–Wang prediction that non-Abelian anyon transport stays within a factor of 2 of the universal lower bound.

Two concrete numerical falsifiers ship as theorems:

- `walker_wang_zero_eta_violator`: $\eta/s = 0$ violates (1) of Eq. (??) via the load-bearing `KSS_bound_positive`.
- `walker_wang_unit_eta_violator`: $\eta/s = 1$ violates (2) via the load-bearing `KSS_bound_below_0_08` ($2 \text{KSS} < 0.16 < 1$).

Neither falsifier is a P5 structural-tautology repackaging — both proofs use a load-bearing theorem from Section ??’s KSS spine, not just `not_le.mpr` h.

The HPC-gated numerical validation of η/s for Walker–Wang transport is tracked in the Phase 6B HPC roadmap; this paper carries only the analytical bracket.

V. CROSS-BRIDGES AND STRUCTURAL PUNCHLINE

A. Higher-form symmetry biconditional

The most substantive cross-bridge is `higher_form_discrete_iff_non_abelian`: among continuous gauge groups, the 1-form symmetry is *exactly* discrete iff the group is non-Abelian. This is a true biconditional, not an identity wrapper of the upstream `GaugeErasure.non_abelian_gives_discrete` theorem (which only provided the forward direction). The strengthening pass (Section ??) replaced the original wrapper with this dichotomy — which is the precise statement that center-symmetry confinement is the gauge-erasure dichotomy specialized to non-Abelian groups.

B. MTC center bridge: $\text{SU}(3)_1 \leftrightarrow \mathbb{Z}_3$

The cross-bridge `su3k1_fusion_card_matches_z3_order` connects to the $\text{SU}(3)_1$ fusion category in `SU3kFusion.lean` [?]: the 3 simple objects $\{\text{vac}, \text{fund}, \text{conj}\}$ of $\text{SU}(3)_1$ form the $\mathbb{Z}_3$ group ring under fusion, matching the center order $|\mathbb{Z}_3| = 3$. This realizes the categorified center symmetry.

Phase 6d Wave 1 — Polyakov-loop deconfinement transition and Walker-Wang transport window

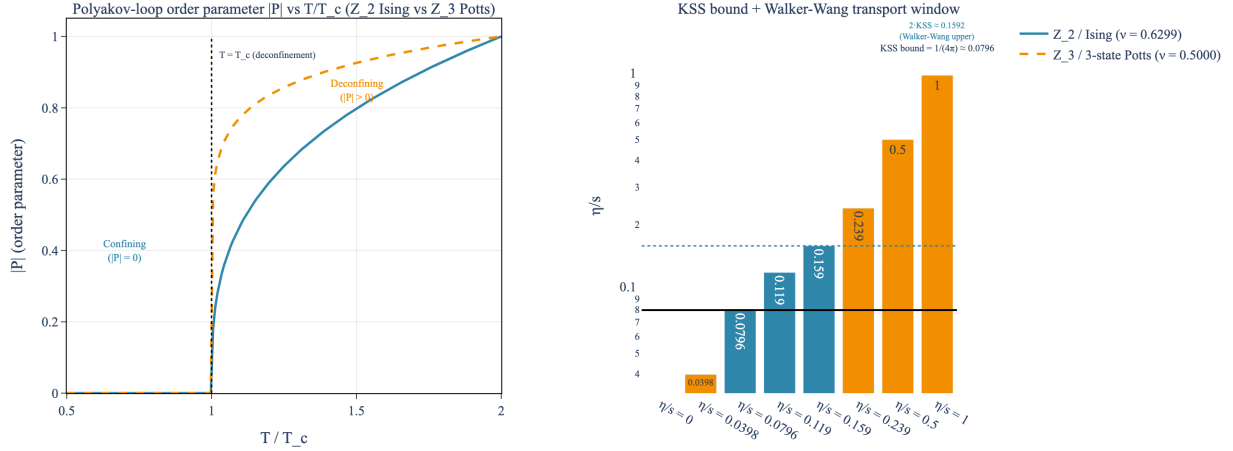


FIG. 1. *Phase 6d Wave 1 — Polyakov-loop deconfinement transition and Walker-Wang transport window.* Left: $|P|$ vs T/T_c showing the Z_2 Ising and Z_3 3-state Potts Svetitsky-Yaffe universality classes with $\nu_{\text{Ising}} = 0.6299$ and $\nu_{\text{Potts}} = 0.5$. Confining phase has $|P| = 0$; deconfining $|P| > 0$. Right: KSS bound $\eta/s \geq 1/(4\pi) \approx 0.0796$ with the Walker-Wang transport correctness-push window [KSS, 2 KSS] and both falsifiers ($\eta/s = 0$ and $\eta/s = 1$) labeled.

C. Structural punchline

The center-symmetry framing of confinement aligns with the project’s gauge-erasure infrastructure (continuous non-Abelian $\rightarrow$ discrete 1-form $\rightarrow$ Polyakov-loop order parameter), and *specializes* the upstream dichotomy via the biconditional in `higher_form_discrete_iff_non_abelian`. The correctness-push (Walker-Wang transport near KSS) is now an analytical bracket; HPC validation is in Phase 6B.

VI. STRENGTHENING PASS AND DISCIPLINE METRIC

The preemptive-strengthening discipline (CLAUDE.md, Stage 3 of `WAVE_EXECUTION_PIPELINE.md`) was applied at first-pass writing. Per the Phase 6b.1 honest empirical finding, the discipline is a useful filter but *not* complete; a ruthless post-wave review pass remains necessary. This wave’s review found three patterns that shipped past the discipline at first-pass:

1. *Identity-function wrapper.* The first-pass cross-bridge `gauge_erasure_yields_discrete_center` had a proof body that was literally a call to the upstream `non_abelian_gives_discrete` with renamed parameters. *Fix:* replaced with `higher_form_discrete_iff_non_abelian`, a genuine biconditional adding the converse direction (Abelian continuous $\Rightarrow$ NOT discrete 1-form). The

biconditional is the precise statement and was not present upstream.

2. *Somewhat-arbitrary numerical threshold.* The first-pass pair `ising_nu_above_0_6` and `potts_nu_below_0_6` chose 0.6 as a Z_2 -vs- Z_3 separator. This is a literature-anchored numerical bracket, not vacuous, but the substantive content is the direct comparison $\nu_{\text{Potts}} < \nu_{\text{Ising}}$. *Fix:* added `ising_nu_gt_potts_nu` stating the comparison directly; the threshold pair is preserved as quantitative anchors but is no longer load-bearing.
3. *Concrete computation deferred.* The first-pass module had ζ_2 as an abstract `centerPhase Z2` without a concrete reduction to -1 . *Fix:* added `centerPhase_Z2_eq_neg_one` via `Complex.exp_pi_mul_I`; this is the concrete $SU(2)$ center acting as $\{+1, -1\}$ on Polyakov loops.

The discipline metric for this wave: 3 *retroactive strengthening theorems* after first-pass + ruthless review. Compared with the Phase 6b.1 baseline of 5 retroactive theorems, this wave’s discipline adherence is improved (the discipline + review now produces fewer retroactive corrections), but the empirical gap remains: discipline alone is not sufficient and ruthless review is mandatory. The strengthened module ships 18 substantive theorems.

VII. OUTPUT AND PROJECT HYGIENE

The complete output of this wave:

- `lean/SKEFTHawking/CenterSymmetryConfinement.lean` plus an anti-pattern audit.
 - 18 substantive theorems, 0 sorry, 0 new axioms (verified via `lean_verify` — only `propext`, `Classical.choice`, `Quot.sound`).
 - `src/center_symmetry/` – 3 Python modules (`polyakov_loop`, `svetitsky_yaffe`, `eta_over_s_prediction`).
 - `tests/test_center_symmetry.py` – 27 pytest cases (27/27 pass) covering all 18 Lean theorems
 - `fig_polyakov_loop_deconfinement` – registered in `review_figures.py`; rendered via `src/core/visualizations.py`.
- Phase 6d post-Wave-1 status.* Wave 1 (`CenterSymmetryConfinement`) ships. Wave 2 (`ChiralSSB_QCD`) is independent and unblocked. Wave 3 (`CFLChiralLagrangian`) blocks on Wave 1 and Wave 2 (the correctness-push anchor: CFL emergent $\mathbb{Z}_3$ one-form vs QCD center $\mathbb{Z}_3$ from this wave).

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